

Software Requirements Specification

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****Software Requirements Specification (SRS)****

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1. Introduction

This Software Requirements Specification (SRS) document describes the requirements for the B09 application, a web and mobile-based platform that enables users to engage in eco-friendly activities, complete quests, and earn badges. The system consists of a backend API, a web frontend, and a mobile application.

2. Purpose

The purpose of this SRS is to define the functional and non-functional requirements of the B09 application, ensuring that all stakeholders have a clear understanding of the system's capabilities, limitations, and expectations.

3. Scope

The scope of this SRS includes the following components:

- * Backend API (server.js)
- * Web frontend (ActivitiesPage.jsx)
- * Mobile application (Activities.tsx)
- * Database models (Activity.js, Badge.js, User.js, Image.js)
- * Routes (userRoutes.js, activityRoutes.js, summarizeRoutes.js, leaderboardRoutes.js)

4. System Overview

The B09 application is designed to provide a user-friendly platform for users to engage in eco-friendly activities, complete quests, and earn badges. The system consists of the following components:

- * Backend API: Handles API calls, database interactions, and routing.
- * Web frontend: Provides a user interface for users to interact with the system.
- * Mobile application: Offers a mobile-based interface for users to access the system.
- * Database: Stores user data, activity data, badge data, and image data.

5. Functional Requirements

The following functional requirements are specified for the B09 application:

- * **User Registration***: The system shall allow users to register and create an account.
- * **Activity Creation***: The system shall enable administrators to create and manage activities.
- * **Quest Completion***: The system shall allow users to complete quests and earn badges.
- * **Badge Earning***: The system shall award badges to users upon completing quests.
- * **Activity Tracking***: The system shall track user activity and progress.
- * **Leaderboard***: The system shall display a leaderboard showcasing top-performing users.
- * **API Calls***: The system shall handle API calls for activities, user registration, and badge earning.

6. Non-Functional Requirements

The following non-functional requirements are specified for the B09 application:

- * **Performance***: The system shall respond to user interactions within 2 seconds.
- * **Security***: The system shall ensure user data is encrypted and protected.
- * **Usability***: The system shall provide an intuitive and user-friendly interface.
- * **Scalability***: The system shall be able to handle a minimum of 1000 concurrent users.
- * **Availability***: The system shall be available 24/7, with a minimum uptime of 99.9%.

7. Architecture

The B09 application shall consist of the following architecture:

- * **Backend***: Built using Node.js, Express.js, and MongoDB.
- * **Web Frontend***: Built using React.js and TypeScript.
- * **Mobile Application***: Built using React Native and TypeScript.
- * **Database***: MongoDB.

8. Risks

The following risks are identified for the B09 application:

* **Technical Debt**: The system may accrue technical debt due to rapid development and deployment.

* **Security Vulnerabilities**: The system may be vulnerable to security threats if not properly secured.

* **User Adoption**: The system may not gain traction with users if the interface is not intuitive or engaging.

9. Constraints

The following constraints are applied to the B09 application:

* **Timeline**: The system shall be developed and deployed within 6 months.

* **Budget**: The system shall be developed and deployed within a budget of \$100,000.

* **Resources**: The system shall be developed and deployed with a team of 5 developers and 2 designers.

10. Future Scope

The following future scope is envisioned for the B09 application:

* **Gamification**: Integrate gamification elements to enhance user engagement.

* **Social Sharing**: Enable users to share their progress and achievements on social media.

* **Partnerships**: Establish partnerships with eco-friendly organizations to offer exclusive content and rewards.

* **Machine Learning**: Integrate machine learning algorithms to personalize user experiences and offer tailored recommendations.

By following this SRS, the development team shall ensure that the B09 application meets the required functional and non-functional requirements, and is developed within the specified constraints and timeline.